

YouTube Video RAG (Retrieval-Augmented Generation) Project Report

Abstract

This project builds a Retrieval-Augmented Generation (RAG) application that answers user questions from a YouTube AI video. The system extracts the transcript, splits it into chunks, creates embeddings, stores them in a FAISS vector database, retrieves relevant chunks for a user query, and generates answers using a Large Language Model.

Objectives

Develop an AI-powered question answering system for YouTube videos; reduce manual searching; provide semantic search over transcripts; demonstrate RAG using Python.

Technology Stack

Python, YouTube Transcript API, LangChain, FAISS, Sentence Transformers/OpenAI Embeddings, Transformers, Streamlit, dotenv.

Workflow

1. User enters a YouTube URL. 2. Transcript is extracted. 3. Text is cleaned and chunked. 4. Embeddings are generated. 5. Embeddings are stored in FAISS. 6. User asks a question. 7. Relevant chunks are retrieved. 8. LLM generates the final answer.

Modules

Transcript Extraction, Text Chunking, Embedding Generation, Vector Database, Retriever, LLM Response Generator, Streamlit Interface.

Advantages

Fast semantic search, improved answer quality, scalable, reusable for lectures, tutorials and AI videos.

Limitations

Requires transcript availability; answer quality depends on transcript quality and embedding model.

Applications

Education, interview preparation, research, lecture summarization, knowledge retrieval, corporate training.

Future Scope

Support multiple videos, multilingual transcripts, PDF/audio ingestion, chat history, cloud deployment.

Conclusion

The project demonstrates how RAG enhances question answering by combining information retrieval with generative AI.